



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 • STANDARD 6.NOS.9

# Represent Polygons on the Coordinate Plane

Lesson 7-7 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.

**Language Objective:** I can explain how I found a distance using the words vertices, coordinates, and absolute value.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

vertex

polygon

perimeter

distance between vertices

scale

Side length



Tap any word to see its meaning and a picture.

1 A corner point of a polygon, named by an ordered pair, is a

2 A closed figure made of line segments is a

3 The total distance around a polygon is its

4 When two vertices share a coordinate, subtract the other and take the

5 What one grid unit stands for in the real situation is the

6

The distance between two joined vertices is a .

## Write About the Math

**Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?**

*¿A dónde van los otros dos vértices — y cómo trabajan juntas las coordenadas conocidas, la escala y el área para decírtelo?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **vertex**  
*vértice*

☐ **polygon**  
*polígono*

☐ **perimeter**  
*perímetro*

☐ **distance between vertices**  
*distancia entre vértices*

☐ **scale**  
*escala*

**Model:** When two vertices share the same *y*-coordinate, the segment between them is horizontal, so its length comes from the *DIFFERENCE* in the coordinate that changes — the *x*-coordinates. — A strong answer explains that the *SHARED* coordinate tells you the direction of the side, and the *DIFFERING* coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The park's width is \_\_\_\_ feet. Using the area formula, its length is \_\_\_\_ feet, which is \_\_\_\_ units on the map — so the new vertices sit that many units below the known ones.
- My answer is \_\_\_\_ because \_\_\_\_.

*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

---

---

---

---

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.

*Mi afirmación es \_\_\_\_.*

- **My evidence is** \_\_\_\_, **so** \_\_\_\_.

*Mi evidencia es \_\_\_\_, entonces \_\_\_\_.*

---

---

---

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

---

---

---

---

---

---

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

---

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

---

## Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Notes 6:** Side length
7. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from  $y = 3$  down to  $y = -3.75$  the distance is  $3 + 3.75 = 6.75$  units, not  $3.75 - 3 = 0.75$ . Distances that cross zero are the whole reason the absolute value is there.*
8. **Try It 1:**
9. **Try It 2:**